

The New Methods III

Scaling Up: The STEP + COAD Iterative Update Framework

VSEPR-SIM thenewmethods

Each element is a bead with full state. Everything evolves from that state. No shortcuts, no magic constants pretending to be physics.

Bead State and System State

$$Z_i = \begin{bmatrix} \mathbf{x}_i \\ \mathbf{v}_i \\ s_i \\ q_i \\ m_i \\ \varepsilon_i \end{bmatrix} \quad \mathbf{Z} = \{Z_1, Z_2, \dots, Z_N\}$$

$\mathbf{x}_i$ = position, $\mathbf{v}_i$ = velocity, s_i = species, q_i = charge/electronic state, m_i = mass, ε_i = internal energy.

Core Update Equation

At iteration n , define the state and two driver vectors:

$$Z^{(n)} = \begin{bmatrix} \mathbf{x}^{(n)} \\ \mathbf{v}^{(n)} \\ T^{(n)} \\ P^{(n)} \\ s^{(n)} \\ q^{(n)} \\ \varepsilon^{(n)} \end{bmatrix} \quad \text{STEP}^{(n)} = \begin{bmatrix} S^{(n)} \\ T^{(n)} \\ E^{(n)} \\ P^{(n)} \end{bmatrix} \quad \text{COAD}^{(n)} = \begin{bmatrix} C^{(n)} \\ O^{(n)} \\ A^{(n)} \\ D^{(n)} \end{bmatrix}$$

Components: S = state descriptor, T = temperature, E = environmental forcing, P = pressure; C = charge state, O = orbital/electronic character, A = accessibility/reactivity access, D = distribution/arrangement.

General update:

$$\boxed{Z^{(n+1)} = Z^{(n)} + \Delta Z_{\text{STEP}_r}^{(n)} + \Delta Z_{\text{COAD}_r}^{(n)}}$$

or compactly:

$$Z^{(n+1)} = Z^{(n)} + F(\text{STEP}^{(n)}, \text{COAD}^{(n)}) \Delta t$$

Reduced Scalar Drivers

$$\text{STEP}_r = w_S S + w_T T + w_E E + w_P P \quad \text{COAD}_r = w_C C + w_O O + w_A A + w_D D$$

Total driver:

$$\Psi = \alpha \text{STEP}_r + \beta \text{COAD}_r$$

Use Ψ to drive mobility, force scaling, energy, conversion, update probability:

$$M^{(n+1)} = M^{(n)} + \eta_M \Psi^{(n)} \quad \varepsilon^{(n+1)} = \varepsilon^{(n)} + \eta_\varepsilon \Psi^{(n)} \quad Y_B^{(n+1)} = Y_B^{(n)} + \eta_Y \Psi^{(n)}$$

STEP and COAD Contribution Vectors

$$\Delta Z_{\text{STEP}_r}^{(n)} = \begin{bmatrix} \alpha_x \nabla P \\ \alpha_v T^{(n)} \\ \alpha_T E^{(n)} \\ \alpha_P \\ 0 \\ 0 \\ \alpha_\varepsilon T^{(n)} \end{bmatrix} \quad \Delta Z_{\text{COAD}_r}^{(n)} = \begin{bmatrix} 0 \\ \beta_v A^{(n)} \\ 0 \\ 0 \\ \beta_s C^{(n)} \\ \beta_q O^{(n)} \\ \beta_\varepsilon D^{(n)} \end{bmatrix}$$

STEP handles macro thermal-pressure-environment forcing; COAD handles electronic, accessibility, and distribution corrections.

Table A – Simple Structural Settling

One species, mild environment, no chemical conversion, no significant thermal change.

n	S	T (K)	E	P	C	O	A	D	Trend	Note
0	initial lattice	300	dry	1.0	0.00	base	0.20	1.00	stable	reference
1	slight shift	300	dry	1.0	0.00	base	0.22	0.98	mild relax	reposition
2	aligned	301	dry	1.0	0.00	base	0.21	0.97	damping	less motion
3	near stable	300	dry	1.0	0.00	base	0.20	0.97	converging	residual small
4	stable	300	dry	1.0	0.00	base	0.20	0.97	converged	stop

Scalar settling example. Let activity metric $R^{(n)}$ evolve as:

$$R^{(n+1)} = R^{(n)} + \alpha \text{STEP}_r^{(n)} + \beta \text{COAD}_r^{(n)}$$

n	$R^{(n)}$	STEP contrib.	COAD contrib.	$R^{(n+1)}$
0	1.000	−0.050	+0.010	0.960
1	0.960	−0.030	+0.005	0.935
2	0.935	−0.015	+0.003	0.923
3	0.923	−0.006	+0.001	0.918

Table B – Thermal Iteration

Thermally activated response: vacancy growth, expansion, mobility change.

Mobility and defect fraction evolve as:

$$M^{(n)} = M_0 + \alpha_T T^{(n)} + \alpha_E E^{(n)} + \beta_A A^{(n)} \quad f_v^{(n+1)} = f_v^{(n)} + k_T T^{(n)} + k_P P^{(n)} - k_R R^{(n)}$$

n	State S	T (K)	E	P	C	O	A	M	f_v	Comment
0	solid ordered	300	inert	1.0	0.00	ground	0.18	0.20	0.0010	baseline
1	warmed	450	inert	1.0	0.00	ground	0.22	0.35	0.0030	defects begin
2	active	650	inert	1.0	0.00	excited mix	0.30	0.60	0.0080	mobility up
3	hot lattice	850	inert	1.0	0.00	excited	0.42	0.95	0.0180	rearrangement
4	near transition	1000	inert	1.0	0.00	excited	0.55	1.25	0.0350	high disorder

Table C – Chemical Iteration

Conversion / reactive evolution. Let $Y_A^{(n)}$ = reactant fraction, $Y_B^{(n)}$ = product fraction.

$$Y_A^{(n+1)} = Y_A^{(n)} - \Delta Y^{(n)} \quad Y_B^{(n+1)} = Y_B^{(n)} + \Delta Y^{(n)} \quad \Delta Y^{(n)} = k_1 \text{STEP}_r^{(n)} + k_2 \text{COAD}_r^{(n)}$$

where:

$$\text{STEP}_r^{(n)} = w_S S^{(n)} + w_T T^{(n)} + w_E E^{(n)} + w_P P^{(n)} \quad \text{COAD}_r^{(n)} = w_C C^{(n)} + w_O O^{(n)} + w_A A^{(n)} + w_D D^{(n)}$$

n	S	T	E	P	C	O	A	Y_A	Y_B	Tendency	Note
0	reactant-rich	300	mild	1.0	0.05	low	0.10	1.00	0.00	very low	initial
1	activated	450	mild	1.0	0.08	mod	0.18	0.94	0.06	low	thermal act.
2	reactive	600	catalytic	1.1	0.12	mod-high	0.30	0.80	0.20	moderate	onset
3	progressing	750	catalytic	1.1	0.15	high	0.48	0.56	0.44	high	formation
4	product-rich	820	catalytic	1.1	0.16	high	0.60	0.32	0.68	high	late-stage
5	near complete	850	catalytic	1.1	0.15	high	0.63	0.18	0.82	moderate	depleting

Compact reaction-rate form. Define reaction score $R_\chi^{(n)} = \alpha \text{STEP}_r^{(n)} + \beta \text{COAD}_r^{(n)}$, then:

$$Y_A^{(n+1)} = Y_A^{(n)} - \gamma R_\chi^{(n)} \quad Y_B^{(n+1)} = Y_B^{(n)} + \gamma R_\chi^{(n)}$$

Master Combined Table Template

Use this template for any simulation report or note.

n	Phase	S	T	E	P	C	O	A	D	Ψ	E	Conv.
0	initial	base	300	dry	1.0	0.00	base	0.10	1.00	–	1.00	0.00
1	warmed	shift	400	dry	1.0	0.02	base	0.18	0.99	low	1.10	0.03
2	active	reordered	600	catalytic	1.1	0.08	raised	0.30	0.95	mod	1.40	0.15
3	reactive	mixed	750	catalytic	1.1	0.14	high	0.47	0.91	high	1.85	0.38
4	evolved	prod-rich	850	catalytic	1.1	0.16	high	0.60	0.87	high	2.05	0.67

Worked Mini Example at Iteration 2

Given: $T = 600$, $E = 1.0$, $P = 1.1$, $C = 0.12$, $A = 0.30$, $D = 0.95$.

With: $\text{STEP}_r = 0.001 T + 0.2 E + 0.1 P$ and $\text{COAD}_r = 0.5 C + 0.4 A + 0.2 D$:

$$\text{STEP}_r = 0.001(600) + 0.2(1.0) + 0.1(1.1) = 0.600 + 0.200 + 0.110 = 0.910$$

$$\text{COAD}_r = 0.5(0.12) + 0.4(0.30) + 0.2(0.95) = 0.060 + 0.120 + 0.190 = 0.370$$

$$\Psi = \text{STEP}_r + \text{COAD}_r = 1.280$$

With conversion update $Y_B^{(n+1)} = Y_B^{(n)} + 0.05 \Psi$ and $Y_B^{(n)} = 0.20$:

$$\Delta Y_B = 0.05 \times 1.280 = 0.064 \quad \implies \quad Y_B^{(n+1)} = 0.264$$

Clean Report-Ready Form

$$Z^{(n+1)} = Z^{(n)} + \Delta Z_{\text{STEP}_r}^{(n)} + \Delta Z_{\text{COAD}_r}^{(n)} \quad (1)$$

$$\text{STEP}_r = w_S S + w_T T + w_E E + w_P P \quad (2)$$

$$\text{COAD}_r = w_C C + w_O O + w_A A + w_D D \quad (3)$$

$$\Psi = \alpha \text{STEP}_r + \beta \text{COAD}_r \quad (4)$$

$$Y^{(n+1)} = Y^{(n)} + \eta \Psi^{(n)} \quad (5)$$

That is the whole engine in compact form.

Table Summary: When to Use Which

Table	Use for
A – Simple structural	Settling, baseline state evolution, elastic relaxation
B – Thermal	Vacancy growth, mobility, phase drift, thermal disorder
C – Chemical	Conversion, accessibility-driven reactivity, product growth